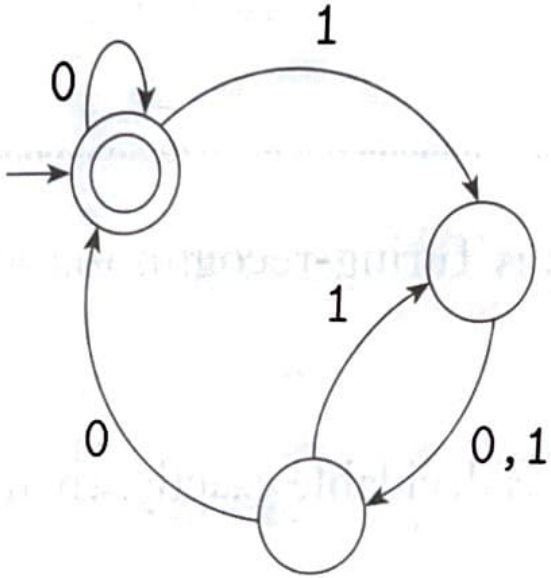
EXERCISES

A4.1 Answer all parts for the following DFA M and give reasons for your answers.



a. Is $\langle M, 0100\rangle \in A_{\mathsf{DFA}}$?

b. Is $\langle M, 011\rangle \in A_{\mathsf{DFA}}$?

c. Is $\langle M\rangle \in A_{\mathsf{DFA}}$?

d. Is $\langle M, 0100\rangle \in A_{\mathsf{REX}}$?

e. Is $\langle M\rangle \in E_{\mathsf{DFA}}$?

f. Is $\langle M, M\rangle \in EQ_{\mathsf{DFA}}$?

4.2 Consider the problem of determining whether a DFA and a regular expression are equivalent. Express this problem as a language and show that it is decidable.

4.3 Let $ALL_{\mathsf{DFA}} = \{\langle A\rangle \mid A \text{ is a DFA and } L(A) = \Sigma^*\}$. Show that ALL_{DFA} is decidable.

4.4 Let $A\varepsilon_{\mathsf{CFG}} = \{\langle G\rangle \mid G \text{ is a CFG that generates } \varepsilon\}$. Show that $A\varepsilon_{\mathsf{CFG}}$ is decidable.

A4.5 Let $E_{\mathsf{TM}} = \{\langle M\rangle \mid M \text{ is a TM and } L(M) = \emptyset\}$. Show that $\overline{E_{\mathsf{TM}}}$, the complement of E_{TM}, is Turing-recognizable.

4.6 Let X be the set $\{1, 2, 3, 4, 5\}$ and Y be the set $\{6, 7, 8, 9, 10\}$. We describe the functions $f\colon X \longrightarrow Y$ and $g\colon X \longrightarrow Y$ in the following tables. Answer each part and give a reason for each negative answer.

4.14 Let $C = \{\langle G, x\rangle \mid G \text{ is a CFG } x \text{ is a substring of some } y \in L(G)\}$. Show that C is decidable. (Hint: An elegant solution to this problem uses the decider for E_{CFG}.)

***4.15** Let $E = \{\langle M\rangle \mid M \text{ is a DFA that accepts some string with more 1s than 0s}\}$. Show that E is decidable. (Hint: Theorems about CFLs are helpful here.)

***4.16** Let $PAL_{\text{DFA}} = \{\langle M\rangle \mid M \text{ is a DFA that accepts some palindrome}\}$. Show that PAL_{DFA} is decidable. (Hint: Theorems about CFLs are helpful here.)

A*4.17 Let $BAL_{\text{DFA}} = \{\langle M\rangle \mid M \text{ is a DFA that accepts some string containing an equal number of 0s and 1s}\}$. Show that BAL_{DFA} is decidable. (Hint: Theorems about CFLs are helpful here.)

4.18 A ***useless state*** in a pushdown automaton is never entered on any input string. Consider the problem of determining whether a pushdown automaton has any useless states. Formulate this problem as a language and show that it is decidable.

A*4.19 Say that an NFA is ***ambiguous*** if it accepts some string along two different computation branches. Let $AMBIG_{\text{NFA}} = \{\langle N\rangle \mid N \text{ is an ambiguous NFA}\}$. Show that $AMBIG_{\text{NFA}}$ is decidable. (Suggestion: One elegant way to solve this problem is to construct a suitable DFA and then run E_{DFA} on it.)

4.20 Let $S = \{\langle M\rangle \mid M \text{ is a DFA that accepts } w^{\mathcal{R}} \text{ whenever it accepts } w\}$. Show that S is decidable.

4.21 Let $PREFIX\text{-}FREE_{\text{REX}} = \{\langle R\rangle \mid R \text{ is a regular expression and } L(R) \text{ is prefix-free}\}$. Show that $PREFIX\text{-}FREE_{\text{REX}}$ is decidable. Why does a similar approach fail to show that $PREFIX\text{-}FREE_{\text{CFG}}$ is decidable?

4.22 Let A and B be two disjoint languages. Say that language C ***separates*** A and B if $A \subseteq C$ and $B \subseteq \overline{C}$. Show that any two disjoint co-Turing-recognizable languages are separable by some decidable language.

***4.23** Prove that the class of decidable languages is not closed under homomorphism.

***4.24** Let C be a language. Prove that C is Turing-recognizable iff a decidable language D exists such that $C = \{x \mid \exists y\ (\langle x, y\rangle \in D)\}$.

4.25 Prove that EQ_{DFA} is decidable by testing the two DFAs on all strings up to a certain size. Calculate a size that works.

4.26 Let $A = \{\langle R\rangle \mid R \text{ is a regular expression describing a language containing at least one string } w \text{ that has } 111 \text{ as a substring (i.e., } w = x111y \text{ for some } x \text{ and } y)\}$. Show that A is decidable.

***4.27** Show that the problem of determining whether a CFG generates all strings in 1^* is decidable. In other words, show that $\{\langle G\rangle \mid G \text{ is a CFG over } \{0,1\} \text{ and } 1^* \subseteq L(G)\}$ is a decidable language.

A4.28 Let $\Sigma = \{0,1\}$. Show that the problem of determining whether a CFG generates some string in 1^* is decidable. In other words, show that

$$\{\langle G\rangle \mid G \text{ is a CFG over } \{0,1\} \text{ and } 1^* \cap L(G) \neq \emptyset\}$$

is a decidable language.

4.29 Let $A = \{\langle R, S\rangle \mid R \text{ and } S \text{ are regular expressions and } L(R) \subseteq L(S)\}$. Show that A is decidable.

A4.30 Let $A = \{\langle M\rangle \mid M \text{ is a DFA that doesn't accept any string containing an odd number of 1s}\}$. Show that A is decidable.

4.31 Let $INFINITE_{\mathsf{PDA}} = \{\langle M \rangle \mid M \text{ is a PDA and } L(M) \text{ is an infinite language}\}$. Show that $INFINITE_{\mathsf{PDA}}$ is decidable.

A4.32 Let $INFINITE_{\mathsf{DFA}} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) \text{ is an infinite language}\}$. Show that $INFINITE_{\mathsf{DFA}}$ is decidable.